# SPLITTING OF THE CONORMAL SEQUENCE FOR A COMPOSITE OF IMMERSIONS DETECTS REGULARITY OF BOTH FACTORS OVER A LOCALLY NOETHERIAN BASE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-extra-permanence-regular-immersion-noetherian is carried in claim_key and here rather than in a per-claim tag URL.

*Editorial note.* The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

*Editorial note.* Statement span 5144-5163 and proof span 5165-5197 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

*Editorial note.* The statement contains a displayed three-term conormal sequence with a split exactness condition; it is copied verbatim from the counted statement span.

*Editorial note.* The proof refers to 'all 4 types of regular immersions' agreeing in the locally Noetherian case; the author's wording is preserved.

*Editorial note.* 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

*Allowed standard prerequisites (editorial).*

- Over a locally Noetherian scheme all four notions of regular immersion agree and regularity can be checked on local rings, cited from the same chapter. Cited at proof lines 5165-5168. Reduces the whole statement to a statement about the local rings at the three points.
- The composite of two regular immersions is a regular immersion, and the transitivity statement relating the conormal sequence of a composite to those of its factors, both cited from the same chapter. Cited at proof lines 5169-5172. Gives the implications from the first condition to the second and from the second to the third.
- Any minimal sequence of generators of a quasi-regular ideal is a quasi-regular, hence regular, sequence, cited from the same chapter. Cited at proof lines 5180-5182. Makes a minimal generating set of the ideal of the composite immersion a regular sequence.
- A minimal generating set of an ideal whose conormal sequence has a splitting compatible with a subideal can be chosen with the subideal's generators first, and a regular sequence splits into a regular sequence and a regular

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sequence modulo its first part. Used at proof lines 5183-5195. Produces the two shorter regular sequences that witness regularity of the individual immersions.

*Source reference.* Original lemma 063I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7832–7832. Same-chapter reference used in the retained author text at source line(s) 4939, 4970, 5169, 5234, 5705. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 067Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7979–7979. Same-chapter reference used in the retained author text at source line(s) 5171, 5669, 5817. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 067N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7977–7977. Same-chapter reference used in the retained author text at source line(s) 5113, 5133, 5183. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 063N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7837–7837. Same-chapter reference used in the retained author text at source line(s) 5173. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

**Lemma 0.1.** *Let $i : Z \to Y$ and $j : Y \to X$ be immersions of schemes. Pick $z \in Z$ and denote $y \in Y$, $x \in X$ the corresponding points. Assume $X$ is locally Noetherian. The following are equivalent*

1. *$i$ is a regular immersion in a neighbourhood of $z$ and $j$ is a regular immersion in a neighbourhood of $y$,*
2. *$i$ and $j \circ i$ are regular immersions in a neighbourhood of $z$,*
3. *$j \circ i$ is a regular immersion in a neighbourhood of $z$ and the conormal sequence*
$$0 \to i^*\mathcal{C}_{Y/X} \to \mathcal{C}_{Z/X} \to \mathcal{C}_{Z/Y} \to 0$$
*is split exact in a neighbourhood of $z$.*

*Proof.* Since $X$ (and hence $Y$) is locally Noetherian all 4 types of regular immersions agree, and moreover we may check whether a morphism is a regular immersion on the level of local rings, see Lemma 063I. The implication (1) $\Rightarrow$ (2) is Lemma 067Q. The implication (2) $\Rightarrow$ (3) is Lemma 063N. Thus it suffices to prove that (3) implies (1).

Assume (3). Set $A = \mathcal{O}_{X,x}$. Denote $I \subset A$ the kernel of the surjective map $\mathcal{O}_{X,x} \to \mathcal{O}_{Y,y}$ and denote $J \subset A$ the kernel of the surjective map $\mathcal{O}_{X,x} \to \mathcal{O}_{Z,z}$. Note that any minimal sequence of elements generating $J$ in $A$ is a quasi-regular hence regular sequence, see Lemma 067N. By assumption the conormal sequence
$$0 \to I/IJ \to J/J^2 \to J/(I + J^2) \to 0$$

is split exact as a sequence of $A/J$-modules. Hence we can pick a minimal system of generators $f_1, \ldots, f_n, g_1, \ldots, g_m$ of $J$ with $f_1, \ldots, f_n \in I$ a minimal system of generators of $I$. As pointed out above $f_1, \ldots, f_n, g_1, \ldots, g_m$ is a regular sequence in $A$. It follows directly from the definition of a regular sequence that $f_1, \ldots, f_n$ is a regular sequence in $A$ and $\overline{g}_1, \ldots, \overline{g}_m$ is a regular sequence in $A/I$. Thus $j$ is a regular immersion at $y$ and $i$ is a regular immersion at $z$. $\square$